

Cosmic Fracture Lines: Mapping the Expansionist vs. Stewardship Divide in AI and Space Governance

Philosophical Foundations: Growth Versus Responsibility

The contemporary discourse surrounding artificial intelligence (AI) and space exploration is underpinned by a fundamental philosophical tension between two divergent worldviews: expansionism and stewardship. This ideological divide is not merely an academic distinction but a practical framework that shapes national policies, corporate strategies, and the very direction of technological progress. Expansionism, most prominently articulated through the lens of techno-optimism, champions rapid, unconstrained technological advancement as the primary engine of human flourishing and survival [5](#) [6](#). In stark contrast, stewardship advocates for a more cautious, value-driven approach that prioritizes long-term sustainability, intergenerational equity, and environmental or cosmic preservation [1](#) [53](#).

The expansionist philosophy, often referred to as techno-optimism, posits that growth is intrinsically linked to progress and vitality, while stagnation is equated with decline and eventual collapse [5](#). Proponents of this view, exemplified by figures like Marc Andreessen, argue that technology is the only perpetual source of growth, capable of creating material abundance and solving all manner of problems [5](#) [9](#). This perspective elevates free markets to the status of a "discovery machine," viewing them as the optimal system for fostering innovation and wealth creation [6](#). Within this framework, AI is celebrated as a universal problem-solving tool and a critical lever for future progress, with any slowdown in its development being framed as a moral failing or even a form of preventable harm [5](#) [9](#). This worldview explicitly rejects concepts traditionally associated with responsible development, such as sustainability, social responsibility, stakeholder capitalism, and the Precautionary Principle, which are dismissed as impediments to progress and "bad ideas" derived from anti-growth ideologies [5](#) [7](#). The philosophical roots of techno-optimism draw heavily from Nietzschean "master morality," where creators and innovators define their own values, and opposition is seen as

stemming from "slave morality"—a position of resentment and fear ⁷. It also shows influence from the Italian Futurist movement, with its glorification of speed, war, and militarism as catalysts for change ⁷. This philosophy fundamentally frames humanity's mission as an expansion into the cosmos, driven by technological prowess and market forces ⁵.

Conversely, the philosophy of stewardship is built upon a foundation of ethical responsibility, caution, and long-term thinking. It redefines humanity's role in the universe not as proprietors but as responsible caretakers of a shared cosmic environment ¹. This perspective emphasizes that our actions have long-term impacts on ecosystems and future generations, necessitating cooperation and prudent management ⁵³. Key principles of stewardship include valuing extraterrestrial environments for their intrinsic worth, not just their extractable resources; ensuring that resource use does not benefit a few at the expense of many or the environment itself; and adhering to a principle of leaving celestial bodies no worse—and perhaps better—than they were found ¹. This worldview inherently demands robust governance, transparency, and accountability, aligning closely with the precautionary principle, which calls for preventative action in the face of uncertain but potentially severe risks ¹¹⁶¹⁵⁷. Stewardship finds resonance in various philosophical traditions, including virtue ethics, which encourages engineers and developers to cultivate traits like honesty and responsibility in their work ⁷¹, and Indigenous philosophies that emphasize respect, reciprocity, and interrelatedness ⁵⁴. It also intersects with theological concepts of stewardship, which frame the management of creation as a moral duty ^{56 132}. In the context of space, stewardship provides a normative framework for planetary protection, sustainable exploration, and the equitable sharing of benefits derived from outer space activities ^{3 196}.

The Lunar Norms War: Artemis Accords vs. Moon Agreement

The ideological conflict between expansionism and stewardship is most vividly illustrated in the realm of international space law, particularly through the divergence between the Artemis Accords and the 1979 Moon Agreement. These two frameworks represent competing visions for the governance of the Moon and other celestial bodies, reflecting the deep-seated philosophical differences regarding property rights, resource utilization, and international cooperation in space. The Artemis Accords, initiated by the United

States, lean towards an expansionist interpretation of existing treaties, while the Moon Agreement embodies a stewardship-oriented model centered on collective ownership and equitable benefit-sharing.

The Artemis Accords, a set of non-binding principles established in 2020, aim to create a consensus-based framework for civil space exploration and use, particularly concerning lunar resources [22 115](#). Initiated by NASA and the U.S. Department of State, the Accords are designed to operationalize the norms laid out in the 1967 Outer Space Treaty (OST) for a new era of commercial activity [78](#). Their key provisions include promoting transparency in space activities, interoperability of systems, and the establishment of "safety zones" around operations to prevent interference [219225](#). A central tenet of the Accords is the commitment to the extraction and use of space resources, such as water ice and minerals, which signatories believe can be done in a manner compliant with the OST [17 75](#). While the Accords also call for the preservation of space heritage, such as historically significant landing sites, their emphasis on resource utilization has drawn criticism [139217](#). The Accords are viewed by some as a significant attempt to establish a new norm for space mining that favors early entrants and private actors, reinforcing a privatized, first-come-first-serve model for resource exploitation [127](#). This approach implicitly contradicts the spirit of the Moon Agreement and has been resisted by nations that prefer to develop new space law through multilateral forums like the UN Committee on the Peaceful Uses of Outer Space (COPUOS) [65](#).

In direct opposition stands the Moon Agreement of 1979, formally known as the Agreement Governing the Activities of States on the Moon and Other Celestial Bodies. This treaty, adopted by the United Nations General Assembly, was drafted within the same UN forum that produced the OST and was initially supported by the United States [92](#). Its foundational principle is that the Moon and its natural resources are the "common heritage of mankind" [87 168](#). This concept dictates that the Moon cannot be appropriated by any single nation and that an international regime must be established to govern the exploitation of its resources, ensuring that the benefits are shared among all countries, regardless of their level of space technology development [243244](#). The agreement thus embodies a strong stewardship philosophy, prioritizing global equity, collective management, and the long-term sustainability of space activities over individual national or corporate claims [197](#). However, despite its progressive ideals, the Moon Agreement has failed to gain widespread acceptance. Major spacefaring nations, including the United States, Russia, and China, have never ratified it, citing concerns that its provisions would hinder commercial development and create an overly restrictive legal regime [92 237](#). The

U.S. government has explicitly rejected the Moon Agreement as irrelevant, further cementing the normative divide in international space law 90 .

Feature	Artemis Accords	Moon Agreement (1979)
Status	Non-binding set of principles / political commitments 79 115	Legally binding international treaty 243
Primary Proponent	United States 115	United Nations General Assembly (originally supported by the U.S.) 92
Core Philosophy	Expansionism / Commercial Utilization 127	Stewardship / Common Heritage 87
Key Provision on Resources	Supports the extraction and use of space resources 75	Declares Moon and its resources the "common heritage of mankind"; requires an international regime for exploitation 168
Legal Basis	Reinforces interpretation of the Outer Space Treaty (1967) 80	A separate treaty intended to clarify and expand upon the OST 88
Focus on Safety	Establishes "safety zones" around operations 225	Focuses on general safety and preservation, but lacks detailed provisions for commercial operations 118
Ratification Status	Signed by 29 nations (as of 2024) 77	Not ratified by any major spacefaring nation (including the U.S., Russia, China); criticized by some as a "treaty in search of a subject" 92 237

This schism creates a significant legal and geopolitical challenge. On one hand, the Artemis Accords provide a flexible, politically driven framework for current exploration efforts, endorsed by a growing coalition of nations and facilitating collaboration on projects like the International Lunar Research Station (ILRS) 64 113. On the other hand, the lack of broad ratification for the Moon Agreement leaves a gap in the legal architecture for long-term resource management and environmental protection, raising concerns about potential conflicts, orbital congestion, and the unregulated exploitation of space 241242. The debate highlights a fundamental disagreement over whether space should be governed by a cooperative commons model or an open-access frontier model, a choice with profound implications for the future of space security and sustainability 84 191.

Governing the Algorithm: Contrasting Global Approaches to AI

The philosophical divide between expansionism and stewardship extends deeply into the domain of artificial intelligence, where it manifests in a fragmented and evolving global landscape of governance. The world's major economic and political blocs—the European

Union, the United States, and others—are pursuing markedly different strategies for regulating AI, reflecting their underlying values regarding innovation, risk, and societal control. These contrasting approaches create a complex patchwork of rules that presents significant challenges for multinational organizations while signaling a broader ideological struggle over the future of technology.

The European Union has adopted a comprehensive, legally binding regulatory approach with the AI Act, which entered into force in August 2024 and becomes applicable in August 2026 [43](#) [93](#). This regulation establishes a risk-based framework that classifies AI systems according to their potential impact. It outright prohibits AI applications deemed to pose "unacceptable risk," such as social scoring by governments and manipulative AI techniques [94](#). For "high-risk" systems, such as those used in critical infrastructure, employment, or law enforcement, the Act imposes strict requirements related to data quality, transparency, human oversight, robustness, and accuracy [100](#). Lower-risk systems may require transparency obligations, while minimal-risk AI is largely left unregulated [99](#). The EU AI Act's design reflects a clear stewardship ethos, aiming to build a trust framework for AI by embedding principles of ethics, lawfulness, and safety directly into the technology lifecycle [98](#). By establishing a uniform legal standard across all member states, the EU is attempting to create a predictable and secure environment for the development and deployment of trustworthy AI [96](#). This proactive, value-oriented model seeks to ensure that AI develops in service of humanity's best interests, guided by the rule of law and ethical principles [192](#).

In contrast, the United States has pursued a less prescriptive, market-oriented strategy for AI governance. The cornerstone of this approach is Executive Order 14110, signed by President Biden in October 2023, which directed federal agencies to develop voluntary guidelines and principles for the safe, secure, and trustworthy development of AI [27](#) [247](#). Rather than creating a single, overarching statute, the order mobilized the development of standards and best practices, with the National Institute of Standards and Technology (NIST) playing a central role [205](#)[246](#). The resulting NIST AI Risk Management Framework (RMF) is a voluntary, non-regulatory guide intended to help organizations manage AI-related risks [205](#). This approach aligns with an expansionist philosophy that prioritizes innovation and avoids burdensome regulations that could hinder American technological leadership [129](#). The subsequent revocation of EO 14110 under a new administration signaled a decisive shift towards removing perceived barriers to AI innovation, further entrenching the preference for a light-touch, guidance-based model [129](#)[260](#). Critics argue that this reliance on voluntary frameworks lacks the enforcement mechanisms and legal

certainty of the EU AI Act, potentially leading to a less coherent and less protective global governance landscape [32](#) [128](#).

Other nations and international bodies are navigating this divide in various ways. China has developed its own regulatory framework, reflecting a state-centric model of governance that differs significantly from both the EU and US approaches [42](#) [46](#). Meanwhile, international organizations like UNESCO and the OECD are working to establish global norms and principles. UNESCO's Recommendation on the Ethics of Artificial Intelligence, adopted in 2021, serves as the first-ever global standard-setting instrument on the topic, emphasizing principles like transparency, fairness, and accountability [72](#) [134](#). Similarly, the OECD AI Principles advocate for responsible stewardship of AI to achieve beneficial outcomes for people and the planet [106](#). These global instruments aim to foster convergence and provide a common language for dialogue, but they lack the legal force of regulations like the EU AI Act. The result is a complex and sometimes contradictory ecosystem of AI governance, where organizations must navigate overlapping and sometimes conflicting requirements from different jurisdictions [47](#) [159](#). This fragmentation underscores the ongoing struggle to balance the imperatives of technological advancement with the need for ethical oversight and public trust.

AI at the Intersection: Dual-Use Technology for Cosmic Ambition

Artificial intelligence has emerged as a powerful dual-use technology at the nexus of space exploration and governance, simultaneously amplifying the ambitions of expansionists and providing essential tools for stewards. Its integration into space systems is accelerating, enabling missions that were previously impossible and creating a host of novel ethical, legal, and strategic challenges. The application of AI in space is not a monolithic trend; rather, it is a contested terrain where the philosophical clash between expansionism and stewardship plays out in real-time, shaping how humanity interacts with the cosmos.

For proponents of expansionism, AI is a critical enabler of their vision for a commercially vibrant and widely inhabited solar system. Autonomy is paramount for this agenda. AI-powered controllers can execute real-world maneuvers on orbiting satellites, allowing spacecraft to fly autonomously without constant input from Earth [33](#). This capability is

indispensable for deep space missions to Mars and beyond, where communication delays make real-time human control impractical [34](#) [233](#). Autonomous navigation allows rovers and landers to explore rugged terrain and make scientific observations independently, vastly increasing the scope and efficiency of exploration [236](#). Furthermore, AI is crucial for the economic ambitions driving expansionism. Autonomous systems can lower the cost and increase the reliability of satellite constellations and future space-based industries [253](#). AI algorithms are being developed to identify and extract valuable resources like lunar water ice, a key component of plans for sustainable, long-term human presence on the Moon [104](#). By optimizing operations and reducing reliance on costly human intervention, AI supports an expansionist model focused on rapid growth, resource acquisition, and the eventual settlement of other worlds [5](#).

However, AI is equally vital for the goals of stewardship, providing the tools necessary to protect the space environment and conduct exploration responsibly. One of the most pressing stewardship concerns is planetary protection—the prevention of both forward contamination (Earth microbes contaminating other worlds) and backward contamination (extraterrestrial materials posing a hazard to Earth) [2](#) [3](#). AI systems can help enforce planetary protection protocols by monitoring spacecraft systems and analyzing data to detect potential breaches, embodying the precautionary principle [116](#)[118](#). Perhaps the most critical stewardship application of AI is in ensuring space sustainability. As the number of objects in orbit grows exponentially, the risk of collisions and the accumulation of space debris increases dramatically [38](#). AI-driven decision support systems are essential for advanced Space Traffic Management (STM), autonomously generating and evaluating collision avoidance maneuvers to protect valuable assets in orbit [35](#) [124](#). The STM market is projected to grow substantially, fueled by the need for AI to manage this orbital congestion [250](#)[262](#). By preventing catastrophic collisions and mitigating debris, AI helps safeguard the space environment for future generations, a core tenet of the stewardship philosophy [240](#).

The dual-use nature of AI in space creates several critical tensions. The most prominent is the trade-off between autonomy and human control. As AI systems take on more complex and high-stakes tasks, questions of accountability become acute [179](#). If an autonomous AI makes a decision that leads to a mission failure or a collision, determining responsibility is challenging [180](#). This tension between the need for rapid, automated decision-making and the ethical requirement for meaningful human oversight is a central dilemma in AI governance for space [190](#). Moreover, the rapid deployment of AI in space is outpacing the development of specific laws and standards, creating a significant governance vacuum [102](#) [121](#). Existing legal frameworks like the Outer Space Treaty were not designed with highly

autonomous, AI-driven systems in mind, leaving critical questions of liability, transparency, and safety unresolved [10](#) . Finally, there are significant security risks. AI-powered space systems can become targets for cyberattacks, which could compromise sensitive data or manipulate autonomous operations, potentially leading to catastrophic failures [40](#) . Securing these systems against malicious actors is a paramount concern for both expansionist and stewardship communities alike.

Policy Imperatives and Strategic Crossroads

The competing philosophies of expansionism and stewardship, manifested in the divergent paths of space law and AI governance, present governments and organizations with a series of critical policy imperatives and strategic decisions. Navigating this complex landscape requires moving beyond reactive measures to proactively shaping the frameworks that will govern humanity's future in space and the technologies that enable it. The choices made today will determine whether space remains a fragile commons or becomes a contested frontier, and whether AI serves as a tool for unchecked growth or a partner in responsible development.

One of the most urgent policy imperatives is the reform and harmonization of international space law. The current legal architecture, primarily based on the 1967 Outer Space Treaty, is increasingly inadequate for addressing modern challenges like commercial resource extraction and AI-driven autonomous operations [76](#) [102](#) . There is a pressing need for new international agreements that can bridge the gap between the expansionist-leaning norms of the Artemis Accords and the stewardship-oriented principles of the Moon Agreement [65](#) . Such an agreement could establish a clear international regime for resource sharing, define technical standards for space traffic management, and codify the responsibilities of operators of autonomous systems. To avoid a chaotic and inefficient regulatory environment, international bodies like UNESCO and the OECD should continue their efforts to promote convergence on ethical AI principles, building upon instruments like the UNESCO Recommendation on AI Ethics to create a more coherent global governance landscape [72](#) [164](#) . Furthermore, policies must mandate robust data stewardship practices for all space-faring entities. This includes ensuring that the vast datasets collected and analyzed by AI systems are managed ethically, securely, and transparently, adhering to principles of accessibility and reusability to maximize scientific benefit while protecting privacy and national security [227231](#) .

For organizations operating in these fields, several strategic crossroads demand careful consideration. The first is the choice of a governance model. Will an organization prioritize agility and market dominance by adopting the US-style approach of voluntary frameworks and self-regulation, or will it opt for risk aversion and long-term stability by aligning with the more prescriptive requirements of the EU AI Act? This decision will profoundly impact its operational costs, legal liabilities, and market access. The second critical choice involves defining "safety." Is safety defined narrowly by mission success and commercial viability, or does it encompass a broader set of considerations, including environmental impact, long-term sustainability, and the equitable distribution of benefits? A narrow definition aligns with an expansionist mindset, while a broad one reflects a stewardship ethos. Third, entities must decide how to engage in international collaboration. Will they focus on bilateral arrangements and domestic policy to gain a competitive advantage, or will they invest in multilateral forums like COPUOS to help shape global norms and mitigate systemic risks? Finally, the balance between autonomy and human oversight must be strategically determined. How much authority will be delegated to AI systems in critical space operations? This decision has profound implications for mission reliability, security, and the allocation of accountability in case of failure [179222](#).

These policy and strategic choices are not abstract exercises; they have tangible consequences. A failure to develop robust international governance could lead to increased geopolitical tensions, orbital congestion, and a degradation of the space environment, hindering exploration for all [241](#). Conversely, a successful effort to build a cooperative, stewardship-based framework could foster a stable and sustainable space economy, unlocking immense scientific and commercial potential while ensuring that the benefits are shared broadly. The path chosen will reflect society's deepest values regarding growth, responsibility, and our place in the cosmos.

Future Trajectories: Scenarios for Humanity's Extraterrestrial Footprint

Synthesizing the ideological fault lines, policy debates, and technological trajectories reveals several plausible future scenarios for humanity's engagement with space, shaped by the enduring tension between expansionist ambition and stewardship responsibility. These scenarios range from a fragmented and conflict-prone frontier to a regulated and cooperative commons, illustrating the high stakes of the decisions being made today. The

ultimate trajectory will depend on the relative influence of expansionist coalitions and stewardship advocates in shaping the next generation of space norms and technologies.

Scenario 1: The Fractured Frontier – An Expansionist Trajectory. In this scenario, the expansionist worldview prevails, leading to a commercially dynamic but geopolitically unstable space environment. The Artemis Accords and their associated norms become the de facto international standard, embraced by a coalition of like-minded nations and corporations. This coalition establishes a precedent for unilateral or bilateral resource extraction, with early-movers claiming and developing resource-rich areas on the Moon and asteroids. Disputes over "safety zones" and resource claims escalate, creating friction with nations outside the coalition who adhere more closely to the traditional interpretations of the Outer Space Treaty [127225](#). AI governance remains weak and fragmented, with corporations deploying highly autonomous systems to maximize efficiency and minimize costs, often with minimal external oversight [128](#). This results in a series of accidents caused by poorly coordinated autonomous operations, highlighting the need for stronger international rules [255](#). While space becomes a hub of economic activity, the lack of a unified governance structure fosters an arms race in space technology and heightens the risk of conflict [234](#). The space environment becomes increasingly congested and polluted, with orbital debris posing a growing threat to all space assets [38](#).

Scenario 2: The Cooperative Commons – A Stewardship Trajectory. This future is characterized by successful international cooperation and the adoption of a robust stewardship framework. Recognizing the limitations of the current legal regime, major spacefaring nations negotiate a new international agreement that incorporates the "common heritage of mankind" principle for space resources [87](#). This treaty establishes a strong international body to oversee resource utilization, ensuring that benefits are shared equitably among all humankind and that environmental protections are strictly enforced [197](#). Concurrently, the EU AI Act serves as a global benchmark for ethical AI, influencing the development of sector-specific regulations for high-stakes domains like space traffic management and defense [43](#). AI in space is deployed under strict oversight, with mandatory human-in-the-loop protocols for critical decisions and transparent algorithms to ensure accountability [10](#). This collaborative approach fosters a stable and secure space environment, enabling large-scale scientific projects and sustainable commercial ventures. The focus shifts from territorial or resource claims to joint exploration and the peaceful use of space, setting a positive precedent for managing shared global commons [84](#). The stewardship model proves to be a reliable foundation for long-term prosperity and security in outer space.

Scenario 3: The Chaotic Commons – A Failure of Governance. In this third scenario, no consensus emerges to resolve the ideological and legal disputes. The Artemis Accords coexist uneasily with the Outer Space Treaty and the legacy of the Moon Agreement, creating a confusing and contested legal landscape ⁶⁵. Competing national and corporate interests pursue their own agendas with little regard for international norms or coordination. Orbital traffic becomes dangerously chaotic, with a lack of effective space traffic management leading to a series of collisions and near-misses ²⁵⁰. The proliferation of autonomous AI systems, deployed without clear rules of engagement or liability frameworks, exacerbates the situation, with incidents ranging from accidental collisions to deliberate cyberattacks on satellites ⁴⁰. The absence of a functioning governance structure leads to a degradation of the space environment, making low Earth orbit increasingly hazardous and hindering future exploration and utilization for everyone. This scenario represents a tragedy of the commons on a cosmic scale, where short-term gains for a few jeopardize the long-term future of space for all humanity.

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